

Task 13 — Link Analysis

Typed hypergraph inference over Form-classes

Event	ITDX26 — U of Arizona Sierra Vista, 8–18 Sept 2026
Track	Algorithm-only, unclassified, laptop-based evaluation
Solicitation	W91RUSI-FCID260001 (SUB-007)
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Handling	UNCLASSIFIED // PROPRIETARY
Version	FormSpace overlay v1.0 — 2026-09-01

1. Overview — Link as Hypergraph Structure over Form-Classes

Link Analysis is the operation of inferring typed relations over Form-classes and evidence items. Nodes in the run graph $G = (V, E, R)$ are Form-classes, devices, sensors, observations, events, hypotheses, COAs, commands, files, calibration records, and models. Edges are typed: measured, produced, derived, corroborated, contradicted, authorized, dispatched, acknowledged, resulted, included. A link survives when it is Form-consistent (source and target belong to compatible equivalence classes under the learned tolerance) and supported by at least two independent nodes across at least two modalities.

2. Method — Typed Relational GNN plus Deterministic Association

Two baselines run before the learned model:

- Deterministic entity/event association: spatiotemporal Mahalanobis distance with position + time + embedding cosine terms.
- Explanation-only PageRank / centrality on the deterministic graph.

The learned model is a typed relational graph neural network with hypergraph incidence H and residual connections; sparse cross-modal attention operates over top- K neighborhoods to prevent over-smoothing on small graphs. Community detection on the same graph gives event-grouping candidates for human adjudication.

3. Inputs

- Rooted observations with hashes and Merkle proofs.
- Emitted Form-class assignments from Task 12.
- Held-out labeled edges from NLM-Bench for link-prediction metrics.
- Ground-truth event graph for controlled stimulus (disclosed).

4. Outputs

- Evidence paths from hazard event to observations, models, and commands.
- Central observations and devices (with rationale).
- Observed-versus-inferred edge toggle for every relation.
- Missing-link recommendations (what evidence would resolve which hypothesis).

- Confidence, provenance, and graph-completeness metrics.

5. Counterfactual Demonstration

Live counterfactual: remove Agaric node from the graph. Report which evidence paths disappear, which communication paths are severed, and which Form-classes lose their supporting relations. This is the operational form of the ablation and it runs deterministically on the replay bundle.

6. Metrics

- Held-out link-prediction AUROC + AUPRC on NLM-Bench (grouped bootstrap, 5 seeds).
- Path recall at k for evidence-path queries.
- Community-detection ARI against ground truth.
- Counterfactual-consistency: predicted vs actual severed paths.
- Graph-completeness percentage.

7. Timing Budget (live, on-site)

- Cold start ≤ 30 s.
- Live inference ≤ 120 s total on the on-site sample.
- Graph update time reported per operation.

8. Failure Modes

- GNN over-smoothing on small graphs — mitigated by residual connections + typed edges; deterministic baseline reported alongside.
- Single-source event with only one modality — automatically downgraded to hypothesis rather than mission-level alert.
- Contradicted edges preserved with negative evidence weight, not deleted.

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